Krishna Sugumar

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ABOUT ME

Mechanical Engineering graduate specializing in vehicle dynamics and chassis development through Formula Student and automotive industry experience. Experienced in suspension and steering design, kinematic analysis, engineering validation, vehicle testing, and simulation-driven development.

EDUCATION AND TRAINING

08/2022 – 05/2026 Chennai, India

BACHELOR OF ENGINEERING (MECHANICAL ENGINEERING) Sathyabama Institute of Science and Technology

Final grade 7.86 | **Thesis** Design and Implementation of an Automated DEF Tank System for Controlled Filling

WORK EXPERIENCE

POWERTRAIN ENGINEERING INTERN – MAHINDRA RESEARCH VALLEY (MAHINDRA & MAHINDRA LTD.) – 02/2026 – 05/2026 – CHENNAI, INDIA

- Contributed to FEAD optimization to reduce accessory drive loads and improve engine efficiency.
- Analyzed Bosch ECU parameters and RWUP data using Concerto, processing and comparing supplier durability datasets with Python and Excel for engineering validation.
- Evaluated supplier durability test cycles to develop representative real-world validation methods.

FORMULA STUDENT – CAPTAIN | VEHICLE DYNAMICS LEAD | FOUNDING MEMBER – HADRON MOTORSPORTS – 12/2022 – 02/2026 – CHENNAI, INDIA

Team Captain

- Led a 40-member multidisciplinary team across design, manufacturing, testing, sponsorship, and competition.
- Coordinated technical activities with six industry sponsors throughout vehicle development.
- Represented the team at SAE Supra 2025 and Formula Bharat 2026, achieving 5th in Engineering Design and 4th in Dynamics.

Vehicle Dynamics Lead

- Led suspension design, defining geometry, hardpoint locations, steering kinematics, and vehicle performance targets.
- Directed suspension, steering, wheel assembly, and overall vehicle integration.
- Reviewed structural validation and implemented lightweight design improvements before manufacturing.
- Led vehicle setup and testing for skidpad, acceleration, slalom, and endurance events.
- Contributed to the team's Overall Championship at Formula Bharat Dynamix 2025 through vehicle dynamics development.

Design Member | Founding Member

- Contributed to the design and development of components across multiple vehicle subsystems.
- Performed structural validation and developed CAD models to support component manufacturing.

IN-PLANT TRAINEE – DAIMLER INDIA COMMERCIAL VEHICLES PVT LTD (DICV) – 05/2025 – 06/2025 – CHENNAI, INDIA

- Worked on simulation-driven product development within the R&D division.
- Supported FEA-based structural assessment and structural simulation of commercial vehicle components.
- Studied simulation-based validation for product development.

QUALITY ENGINEER (INTERN) – MADRAS ENGINEERING INDUSTRIES LTD. (MEI) – 12/2023 – 12/2023 – CHENNAI, INDIA

- Supported inspection and dimensional validation of automotive braking system components.
- Gained practical exposure to automotive manufacturing, quality assurance, and production processes.
- Evaluated the influence of manufacturing tolerances on component quality and functional performance.

PROJECTS

Vehicle Dynamics Package Development for a Formula Student Race Car

- Designed the complete suspension and steering kinematic architecture, defining hardpoints, camber gain, roll centre behaviour, caster, kingpin inclination.
- Performed suspension and steering kinematic analysis and geometry optimization, balancing performance, manufacturability, cost, and reliability.
- Developed wheel assemblies, suspension components, steering assemblies, and chassis interfaces, integrating vehicle subsystems.
- Optimized suspension and steering components for reduced weight while maintaining structural integrity.
- Correlated simulation and tested vehicle test to refine suspension geometry, steering characteristics, and vehicle setup.

06/2025 – 04/2026
Design and Implementation of an Automated DEF Tank System for Controlled Filling

- Designed an automated DEF filling system to improve contamination control and dispensing accuracy.
- Integrated an ESP32-based control unit, TDS sensor, and automated diversion mechanism to verify DEF quality.
- Developed and validated a functional prototype meeting reliability and automotive emission requirements.

07/2024 – 04/2025
Introduction to Advance heat Recovery System for Automobile

- Developed a model to recover waste heat from the exhaust and engine units and convert it into usable electrical energy.
- Integrated TEG modules to convert recovered heat into electrical energy for vehicle electronic systems.

04/2026 – 05/2026
Formula Student Front Suspension Kinematic Analysis and Bump-Steer Optimization Using MSC Adams/Car

- Diagnosed and corrected a bump-steer defect through kinematic optimization, validated under an independent loading condition
- Identified a roll-center migration issue and recommended a spring-rate change to meet industry ride-frequency targets

Full-Vehicle Handling Balance Study — Anti-Roll Bar Distribution and Understeer Gradient Characterization Using IPG CarMaker

- Validated full-vehicle model, a baseline handling characterization (Skid Pad + corner-exit acceleration).
- Quantified before/after result showing how ARB split changed handling balance.

PUBLICATIONS

2026
A Diesel Exhaust Fluid Reservoir

Published Indian Patent Application (Publication No. IN202641066989 A1).

Authors: J. Jayaprabakar; Mano Gnana Sekar G; Krishna S; Solomon N; Sabbareesh S | **Publisher**: Indian Patent Office

SKILLS

Tools

Msc Adams | Optimum K | Matlab | CATIA 3D-Modelling | Solidworks | Python | IPG Car Maker | AVL Concerto

HONOURS AND AWARDS

01/2025
Overall Champion – Formula Bharat Dynamix 2025 – Formula Bharat - Curiosum Tech Pvt Ltd

Overall Champion – Formula Bharat Dynamix 2025

08/2025

SAE Supra 2025 – SAE India

Secured Overall P12 at SAE Supra 2025, with 5th Place in Engineering Design and 4th Place in Business Presentation.

01/2026

Formula Bharat 2026 – Formula Bharat - Curiosum Tech Pvt Ltd

Secured 4th Overall in Dynamics and 11th Overall in Statics at Formula Bharat 2026.

SAE Supra 2026 - Secured Overall rank 4. Secured P2 in Engineering Design. – SAE India